



Risk factors and clinical presentation of cats with feline idiopathic cystitis[☆]

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Feline idiopathic cystitis (FIC) is the most common cause of feline lower urinary tract disease (FLUTD). This retrospective, case-controlled study evaluated possible risk factors associated with FIC and compared different clinical presentations in 64 cats with FIC. Several risk factors known to be involved in FLUTD were identified as playing a role in FIC. Of the stressful situations considered, most did not occur with increased frequency in cats with FIC compared to controls, except for a house move. The presence of pyuria, haematuria and an increased urine protein:creatinine ratio were significantly higher in obstructed males compared with non-obstructed males. An obstruction was significantly more likely in cats with struvite crystalluria compared with cats without struvite crystalluria. These findings suggest that urethral plugs might be an important cause or contributing factor of obstruction in FIC. Episodes of FIC seem to occur mainly in susceptible cats in combination with a deficient environment.

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Feline lower urinary tract disease (FLUTD) is a general term used to describe the various causes of lower urinary tract signs in cats, which include idiopathic, urolithiasis, urinary tract infection, and anatomic and neurological abnormalities. Several studies have found feline idiopathic cystitis (FIC) to be the most common cause of FLUTD. These studies showed that 54–64% of all cats presenting with FLUTD were idiopathic,^{1–5} and 20–55% of those had an urethral obstruction.^{1,5} Urethral plugs were diagnosed in 10–22% of cats with FLUTD,^{1,2,5} and in one study, urethral plugs were found in 59% of obstructed males.¹ A diagnosis of FIC can only be made when an underlying cause for the clinical signs of FLUTD cannot be identified after a proper evaluation.²

The pathogenesis of urethral obstructions in cats with FIC remains unclear. Obstructive FIC may result from inflammation of the urethra, urethral muscular

spasms, neurological dysfunction and intraluminal plug formation. Urethral plugs can consist of an accumulation of inflammatory cells, erythrocytes, and a matrix of proteins and crystals.^{6–8} Several authors have concluded that struvite is the most common type of crystal found in urethral plugs.^{2,9,10} Recent studies have shown that urethral obstruction in male cats is associated with uroliths in 29% of cases, urethral plugs in 18% of cases, and it is idiopathic (without a urethral plug) in 53% of cases.^{5,11}

Several studies have investigated risk factors for FLUTD (which can be regarded as a group of different diseases).^{3–5,12–14} Some of the most important identified risk factors include being overweight, inactive and experiencing several stressful situations. In contrast, only one case-controlled study assessing the risk factors in cats with FIC has been performed.¹⁵ In that study, cats with FIC were significantly more likely to be male, pedigreed, longhaired and overweight. Several possible stress factors for FIC were identified, the most important of which was living in conflict with another cat in the same household. Many other factors were not found to be significantly different between cases and controls.¹⁵

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Because there are few studies evaluating specific risk factors associated with FIC, the first objective in this study was to evaluate possible risk factors associated with FIC. The second objective was to evaluate whether different clinical presentations of FIC (with or without obstruction and with or without struvite crystalluria) are associated with different risk factors.

Materials and methods

A retrospective, case-controlled study of cats with FIC was performed. Medical records were reviewed for all cats that presented with signs of obstructive or non-obstructive FLUTD at the companion animal clinic at the Faculty of Veterinary Medicine of Ghent University (Belgium) between 2002 and 2008. Inclusion criteria for the FIC group included the presence of clinical signs typically associated with FLUTD and the exclusion of other underlying causes of FLUTD based on the results of a physical examination, urinalysis including a culture, and abdominal ultrasound. A diagnosis of FIC was made for cats with a negative urine culture and no evidence of urinary stones or neoplasia based on bladder and proximal urethral ultrasonography. An equal number of cats that had never experienced signs of FLUTD were also evaluated as a control group. The FIC and control groups were matched for gender and age. Cats in the control group were randomly selected from clients, students and friends. A questionnaire was completed by telephone interview of each owner.

The following information was gathered for all cats: breed, coat length, gender, neuter status, body weight, body condition score, age at first episode of cystitis, food (dry/canned), water intake, use of a litter box, number and location of litter boxes, sleeping location, access to the outside, activity level, hunting behaviour, number of cats in the household, interaction with outdoor cats, general behaviour of the cat (nervous, fearful or aggressive), reaction towards unknown visitors in the house, presence of conflict between cats, position in conflict (submissive, neutral or dominant), occurrence of specific stressful situations during a 3-month period prior to the first episode of cystitis (building work in/at the house, house move, prolonged absence of primary caretaker, cat on holiday/boarding, introduction of new animals or baby, or other stress factors the owner could think of). Additional information from the cats with FIC was gathered as follows: duration of hospitalisation, clinical signs, duration and season of episodes, results of urinalysis and ultrasound examination, concurrent medical illnesses, management, total number of episodes, interval between episodes, relapses, life status and cause of death.

First, all cats with FIC were compared with the controls for all the variables. Then, male cats with FIC that had at least one urethral obstruction were compared with male cats with FIC without any episodes of obstruction. For this analysis only a limited number

of variables were compared: body weight, age, food, access to the outside, total number of episodes and interval between episodes. Specifically for the results of urinalysis and ultrasound examination, comparisons were made between male cats with obstruction at presentation and male cats without obstruction at presentation. Finally, comparisons were made between cats with FIC with and without struvite crystalluria. The same limited number of variables was compared as above, as well as the presence or absence of obstruction at presentation.

Urine was collected by cystocentesis or urinary catheterisation. Within 30 min of urine collection, microscopic evaluation of urinary sediment was performed by a clinician onsite, followed by a semi-quantitative urinalysis and urine culture. Urine protein:creatinine ratios (UPCs) above 0.4 were considered to be elevated. Haematuria or pyuria were considered present when more than 25 red or white blood cells/ μ l were counted. An automated urine analyser performed cell counts by flow cytometry (UF 100, Sysmex). The number of red or white blood cells per high power field multiplied by a conversion factor of 5.5 is comparable with the number of cells per μ l.

For statistical analysis, the Fisher exact test was used for binary variables and nominal categorical variables, the Wilcoxon rank sum test was used for ordinal categorical variables and the *t*-test was used for continuous variables. Differences were considered significant when $P < 0.05$. Percentages were measured based on the number of cats with available or applicable data.

Results

Medical records from 179 cats with possible signs of FLUTD (dysuria, stranguria, pollakiuria, haematuria and periuria) were reviewed, resulting in the inclusion of 64 cats with FIC in the study.

Comparison between the FIC group and the control group

The predominant breed of cats with FIC was the domestic shorthair or longhair ($n = 51$). Persian ($n = 4$), British Shorthair ($n = 3$), Norwegian Forest ($n = 2$), Oriental Shorthair ($n = 1$), Maine Coon ($n = 1$), Russian Blue ($n = 1$) and Ragdoll breeds ($n = 1$) were also represented in the FIC group. No significant breed predispositions were found ($P = 0.234$). The Persian breed was not represented in the control group. Cats with FIC had a significantly ($P = 0.004$) higher body weight ($5.5 \text{ kg} \pm 1.5$) than the control cats ($4.7 \text{ kg} \pm 1.5$). The mean number of cats in the household of a cat with FIC (2.6 ± 2.5) was significantly ($P = 0.044$) higher compared with the control cats (1.9 ± 1.2).

Results for possible risk factors are listed in Table 1. Cats with FIC were significantly more likely to use a litter box ($P < 0.001$), had a lower water intake

Table 1. Possible risk factors for FIC and frequencies in the FIC and control groups.

Variables	Category	FIC	Controls	P	Variables	Category	FIC	Controls	P
Purebred	Yes	13	10	0.234	Long haired	Yes	17	15	0.839
	No	51	54			No	47	49	
Neutered	Yes	62	62	1.000	Use of a litter box	Yes	57	38	<0.001
	No	2	2			No	7	26	
Litter box in a quiet location	Yes	39	29	1.000	Fewer litter boxes than cats	Yes	28	16	0.339
	No	11	8			No	6	7	
Sleeping inside	Yes	61	54	0.076	Multiple cats in household	Yes	37	35	0.859
	No	3	10			No	27	29	
Interaction between single- housed cats and other cats outside	Yes	14	23	0.136	Being more nervous than other cats in the same household	Yes	14	1	<0.001
	No	10	6			No	23	34	
Being more fearful than other cats in the same household	Yes	16	1	<0.001	Being more aggressive than other cats in the same household	Yes	3	4	0.707
	No	21	34			No	34	31	
Conflict with other cat(s)	Yes	35	31	0.119	Conflict with other cat(s) in the same household	Yes	22	18	0.234
	No	16	27			No	29	40	
Conflict with other cat(s) from outside	Yes	14	13	0.657	Occurrence of at least 1 stressful situation from the list mentioned below	Yes	25	20	0.459
	No	37	45			No	39	44	
<i>List of stressful situations:</i>									
Work in or at house	Yes	1	7	0.062	House move	Yes	9	0	0.003
	No	63	57			No	55	64	
Prolonged absence of primary caretaker	Yes	4	0	0.119	Cat on holiday or boarding	Yes	3	2	1.000
	No	60	64			No	61	62	
Introduction of new animals	Yes	2	5	0.439	Introduction of baby	Yes	2	3	1.000
	No	62	59			No	62	61	
Other stress factors the owner can think of	Yes	7	3	0.324					
	No	57	61						
Water intake	Low	22	6	<0.001	Activity level	Low	28	13	<0.001
	Normal	21	44			Normal	32	30	
	High	4	14			High	4	21	
Hunting behavior	Never	11	5	0.006	Reaction toward unknown visitors in the house	Hiding	29	18	0.002
	Sometimes	6	16			Neutral	9	21	
	Often	8	27			Contact	10	25	
Position in conflict	Submissive	11	10	0.055	Body condition	Skinny	1	0	0.008

(continued on next page)

Table 1 (continued)

Variables	Category	FIC	Controls	P	Variables	Category	FIC	Controls	P
Food	Neutral	8	6	0.606	Normal		26	39	<0.001
	Dominant	4	15		Fat		23	23	
	Only dry	31	24		Obese		14	2	
	Mostly dry	7	16		Only inside		34	16	
	Both	25	23		Mostly inside		17	15	
	Mostly canned	0	0		Both		13	18	
	Only canned	1	1		Mostly outside		0	13	
					Only outside		0	2	

($P < 0.001$), had a lower activity level ($P < 0.001$), exhibited less hunting behaviour ($P = 0.006$), had a higher body condition score ($P = 0.008$), and had less access to the outside ($P < 0.001$). Cats with FIC were also characterised as being significantly more nervous ($P < 0.001$) and fearful ($P < 0.001$) than other cats in the same household and were more prone to hide from unknown visitors in the house ($P = 0.002$). The only specific stressful situation identified as significantly more likely in cats with FIC was a house move ($P = 0.003$).

Cats that were purebred, longhaired, or neutered were not significantly more likely to have FIC than the controls. Having fewer litter boxes than the number of cats in the house, sleeping inside, living with other cat(s) in the same household, interaction between single-housed cats and other cats outside, and living in conflict with other cats from the same household or outside were not found to occur significantly more frequently in cats with FIC than in controls. A litter box in a quiet location was not identified as a significant protective factor against FIC. Cats with FIC did not eat a significantly higher proportion of dry food in their diet. No significant differences were found in aggressive behaviour and in the position of cats in conflict. There were not significantly more cats with FIC with at least one specific stressful situation in a 3-month period prior to the first episode when compared with the controls.

Clinical presentations of FIC

Fifty-one male cats (80%) and 13 females (20%) with FIC were included in the analysis. Mean age when the first episode occurred was 50.3 ± 29.8 months. When the first known episode occurred, four cats (6.3%) were younger than 1 year old, 45 (70.3%) were between 1 and 6 years, 13 (20.3%) were between 6 and 10 years, and two (3.1%) were older than 10 years.

During at least one episode, 53 of the FIC cats (83%) had macroscopic haematuria, 45 (70%) had clinical signs of stranguria and periuria, 49 (77%) had dysuria, 50 (78%) had pollakiuria and 37 (58%) had signs of obstruction. Median episode duration was 6.5 days (range: 2–90). In 40 cats (67%), episodes were over within 1 week. Fifteen cats (23%) had at least one episode during the winter and spring, and 20 cats (31%) during the summer and autumn. Six (12%) of the FIC cats with multiple episodes only had obstructive episodes, 20 (41%) only had non-obstructive episodes and 23 (47%) had a combination of both. Of the male cats, 37 (73%) had at least one obstructive episode. Of the group of 27 male cats that presented with an obstruction, urethral plugs were diagnosed in 10 (37%). Mean duration of hospitalisation for obstructed cats was 3.6 ± 1.9 days. According to the owners, episodes seemed to have a clear association with a stressful situation for 11 cats (17%). Known stressful situations occurred during a 3-month period prior to the first episode for 25 cats (39%).

Microscopic haematuria was present in 37 (71%), pyuria in 36 (77%), and struvite crystalluria in 31 (48%) of all the male and female cats. The mean number of erythrocytes in urinary sediment was $8502 \pm 18,092/\mu\text{l}$. The mean number of white blood cells in urinary sediment was $112 \pm 118/\mu\text{l}$. The mean UPC was 1.5 ± 3.0 while the UPC was above 0.4 in 32/57 cats (56%). The mean urinary pH was 6.9 ± 0.6 , and the mean urine specific gravity (USG) was 1.038 ± 0.013 .

Fifty-four cats (84%) were treated medically, while 10 cats (16%) also underwent a surgical treatment. Nine cats had a perineal urethrostomy and one cat had a scrotal urethrostomy. After a perineal urethrostomy, one cat still had obstructions. Seven cats (70%) still had non-obstructive signs of FLUTD after surgery. Despite recommendations, 22 owners (37%) discontinued feeding canned food to their cats. Eight owners (14%) did so because their cat refused to eat a canned diet.

Twenty-five cats (39%) presented at the clinic at the time of the first episode. Fifty-six percent of the male cats that presented with an obstruction had not experienced any previous episodes, whereas only 29% of the male cats that presented without an obstruction did not have any previous episodes. Relapses after presentation occurred in 35 cats (56%). Sixty-two percent of all male cats that presented with an obstruction relapsed, whereas 52% of all non-obstructive male cats relapsed. The mean total number of episodes per cat was 5.5 ± 6.9 . In total, 50 cats (78%) had more than one episode, and 74% of all male cats that presented with an obstruction had multiple episodes, compared with 75% of all non-obstructive male cats. The mean interval between episodes was less than 3 months in 25 cats (51%), between 3 and 6 months in 14 cats (29%), between 6 months and 12 months in six cats (12%) and over 1 year in four cats (8%).

Eight cats (12.5%) died as a result of urinary problems. Three of these cats (37.5%) died during a relapse of a urethral obstruction due to the associated metabolic complications, while the other five cats (62.5%) were euthanased because of persisting clinical signs of FLUTD. Three of these five cats never had an obstructive episode.

Comparisons between different clinical presentations of FIC

Presentation with or without obstruction: as shown in Tables 2 and 3, most of the factors compared between male cats with and without an obstruction were not significantly different. Only four urinary factors differed significantly and were more likely in male cats with an obstruction at presentation: pyuria ($P = 0.023$), microscopic haematuria ($P = 0.036$), an increased UPC ($P < 0.001$), and an increased pH ($P = 0.002$).

Presentation with or without struvite crystalluria: results of the comparison between cats with and without struvite crystalluria are listed in Tables 2 and 3. Most of the compared factors were not significantly

different. Two significant differences were found: pyuria ($P = 0.017$) and an obstruction at presentation ($P = 0.022$) were present in a higher number of cats with struvite crystalluria.

Discussion

Comparison between the FIC group and the control group

Although a specific breed of cat was not identified as having a significantly increased risk for FIC, it was remarkable that the Persian cat, which showed an increased risk for FLUTD in two previous studies,^{4,12} was also the most common purebred in our study. Siamese cats, which have been shown to have a decreased risk for FLUTD,^{4,12} were not represented. A larger study group might have detected statistically significant differences. In contrast to the findings of Cameron et al,¹⁵ purebred or longhaired traits were not found to be significant risk factors for FIC in this study.

Several risk factors that are known to be involved in FLUTD,^{4,12,13} such as higher body weight, higher body condition score, use of a litter box, having a lower water intake and a lower activity level, and having less access to the outside, were also identified in the current study. Several risk factors identified for FIC, such as inactivity and indoor housing, are also known risk factors for feline obesity.¹⁶

Only a few possible stress factors were found to be significantly more present in cats with FIC, including restricted access to the outside and a house move. The number of cats in the household was also significantly higher for a cat with FIC. However, most potential stress factors, such as living with at least one other cat in the same household, interaction between single-housed cats and other cats outside, living in conflict with other cats from the same household or from outside, and the occurrence of most specific stressful situations in a 3-month period prior to the first episode of FIC, did not differ significantly between cats with FIC and controls. Identified stressful situations occurred during a 3-month period prior to the first episode in 39% of the cats with FIC, and in 17% of cases, the owners reported that the episodes seemed to have a clear association with a stressful situation. Living in conflict with another cat might have been underestimated, because signs of silent conflicts can be very subtle and easily missed.¹⁷

Heidenberger¹⁸ showed that there are deficiencies in most indoor cat housing. Modern household environments are, therefore, likely to present stressful situations. However, these situations are not associated with episodes of FIC in most cats of the modern general cat population.

Several typical behavioural characteristics were found in cats with FIC. They tended to be more nervous and fearful than other cats in the same household, and showed more hiding behaviour from unknown visitors in the house. When in conflict, cats with FIC were less

Table 2. Differences in frequencies of risk factors between obstructed and non-obstructed male cats and between the presence and absence of struvite crystalluria.

Variables	Category	Obstructed male cats	Non-obstructed male cats	<i>P</i>	Struvite crystalluria	No struvite crystalluria	<i>P</i>
Microscopic haematuria	Yes	20	10	0.036	20	17	0.541
	No	3	8		6	9	
Pyuria	Yes	19	9	0.023	22	14	0.017
	No	2	8		2	9	
Increased UPC	Yes	22	6	<0.001	18	14	0.289
	No	3	14		10	15	
Urinary sediment on ultrasound	Yes	23	17	1.000	25	26	1.000
	No	1	1		1	1	
Thickening of urinary bladder wall	Yes	15	7	0.089	13	10	0.436
	No	12	17		18	23	
Struvite crystalluria	Yes	18	9	0.051	—	—	—
	No	9	15		—	—	
Obstruction at presentation	Yes	—	—	—	18	9	0.022
	No	—	—		13	24	
Amount of sediment on ultrasound	Small	6	7	0.195	8	9	0.612
	Moderate	5	3		5	5	
	High	10	4		9	7	
Mean interval between episodes	<3 months	18	3	0.501	14	11	0.844
	3–6 months	5	3		6	8	
	6–12 months	3	2		5	1	
	>12 months	4	0		2	2	
Food	Only dry	17	8	0.499	14	17	0.478
	Mostly dry	4	1		3	4	
	Both	15	5		13	12	
	Mostly canned	0	0		0	0	
	Only canned	1	0		1	0	
Access to the outside	Only inside	20	7	0.651	17	17	0.796
	Mostly inside	10	3		8	9	
	Both	7	4		6	7	
	Mostly outside	0	0		0	0	
	Only outside	0	0		0	0	

Presence of microscopic haematuria when more than 25 red blood cells/ μ l in sediment. Presence of pyuria when more than 25 white blood cells/ μ l in sediment. Increased UPC when above 0.4.

likely to be in a dominant position, though this finding did not reach statistical significance. Cats with FIC seem to have less control over their environment, which can be a cause of stress. These findings are compatible with the suggestion that cat-related factors may be important for the expression of FLUTD in indoor-housed cats.¹⁴

Results of this study confirm that several stress factors are present for cats in both the control group and FIC group, and that there are few significant differences in stressors between them. Based on these results and the finding of Heidenberger,¹⁸ a deficient or inadequate

environment alone is not enough to elicit episodes of FIC. Therefore, these results support the statement that episodes of FIC occur mainly in susceptible cats in combination with a deficient environment. This statement is further supported by two other studies. Buffington et al¹⁹ showed a significant reduction in the frequency of lower urinary tract signs when multimodal environmental modification (MEMO) was added to the usual therapy for FIC. Results from another study also suggested a decrease in sickness behaviours in cats with interstitial cystitis when housed in an enriched environment.²⁰ Furthermore, there was

Table 3. Differences in means of continuous risk factors between obstructed and non-obstructed male cats and between the presence and absence of struvite crystalluria.

Variables		Obstructed male cats	Non-obstructed male cats	<i>P</i>	Struvite crystalluria	No struvite crystalluria	<i>P</i>
Body weight (kg)	Mean	5.9	5.4	0.274	5.6	5.4	0.547
	SD	1.6	0.9		1.5	1.5	
Age (months)	Mean	53.4	41.3	0.202	51.4	49.2	0.776
	SD	32.6	20.0		33.2	26.7	
Number of episodes	Mean	5.0	4.0	0.634	5.7	5.2	0.794
	SD	7.2	3.6		7.8	6.1	
Number of red blood cells in urinary sediment/ μ l	Mean	15,460.8	4131.6	0.070	12,080.6	4924.0	0.156
	SD	24,794.9	7970.4		24,036.3	7960.1	
Number of white blood cells in urinary sediment/ μ l	Mean	137.8	97.2	0.340	133.7	88.7	0.196
	SD	139.0	114.2		141.0	86.5	
UPC	Mean	2.4	0.8	0.099	1.7	1.2	0.467
	SD	4.1	1.6		3.9	1.8	
Urinary pH	Mean	7.1	6.7	0.002	7.0	6.8	0.178
	SD	0.5	0.5		0.7	0.5	

SD = standard deviation.

no difference in the mean number of sickness behaviours between cats with interstitial cystitis and healthy cats, when exposed to unusual external events in these enriched environments.

Clinical presentations of FIC

Urethral plugs were diagnosed in 37% of all cases of obstruction. Plugs probably will have been present in a higher number of cats, because a diagnosis of urethral plugs can easily be missed, as they can be flushed back into the urinary bladder during catheterisation.^{5,21}

The presence of pyuria (77%) in the current study was higher compared to a previous study,¹ whereas the presence of struvite crystalluria (48%) was comparable. Because of the retrospective nature of this study and the lack of gradation in the severity of struvite crystalluria in the literature, no further classification besides absence or presence was made. As a large variety in the number of struvite crystals was observed, a more detailed classification would have been interesting. However, the large number of variables influencing the quantitative measurement of crystals makes an accurate estimation of the severity of crystalluria difficult.²² Microscopic evaluation of urinary sediment in search of crystals was always performed by a clinician on site, within 30 min. However, because of the retrospective nature of this study, it cannot be guaranteed that this evaluation was performed consistently within this time limit.

The median duration of an episode (6.5 days) is in agreement with the literature.⁸ However, in 33% of all cats, episodes lasted longer than 1 week, which is longer than the reported 5–7 days.⁸ This could be due to several cats whose clinical signs persisted for a significant period of time (up to 90 days), and referral of those cats is more likely. These results may not reflect the mean duration in the general population.

The high mean total number of episodes (5.5) might not represent the risk of recurrence in the general population because most cats were referral cases. The high number of relapses after presentation (56%) confirms the guarded prognosis found in other studies, which report relapses occurring in 39–66% of all cats with FLUTD.^{5,11,23} Due to the retrospective nature of this study, not all applied therapies could be identified. Increasing water intake was a generally advised therapy, but extensive environmental enrichments were not always discussed with the owners. However, a recent study showed a significant benefit of MEMO in the therapy of FIC.¹⁹ This might have contributed to the high rate of relapses in this study.

Ten cats (16%) underwent surgical intervention for the treatment and/or prevention of urethral obstructions. Repeated obstructions were an important reason to perform surgery. However, cats with and without obstructions had similar relapse rates. Seven cats (70%) still had non-obstructive signs of FLUTD after surgery. Recurrence of FIC was not confirmed by further investigations in most relapses, which is

necessary to rule out the involvement of other causes of FLUTD, such as urinary tract infections. However, these results underline the importance of continuing treatment for FIC after surgery because the underlying disease is not treated with surgery.²⁴

The mortality of FIC in this study (12.5%) was lower than described in a previous study,¹¹ where 26% of all cats with urethral obstruction died because of the disease. It was not the obstruction itself, but elective euthanasia because of persistent obstructive or non-obstructive urinary problems that was the main cause of death in this study.

Comparisons between different clinical presentations of FIC

The presence of pyuria, haematuria, and an increased UPC were significantly higher in male cats with an obstruction when compared with non-obstructed male cats. When compared as continuous variables, there were no significant differences. However, there was a tendency for obstructed males to have a higher number of urinary erythrocytes ($P = 0.070$) and a higher UPC ($P = 0.099$). These findings could possibly be explained by more severe inflammation in obstructed cases. Trauma caused by catheterisation or vesical overdistention associated with the obstruction could potentially contribute to these findings.

Urinary pH was significantly higher in male cats with an obstruction. The hypothesis that plasma proteins leaking into urine during inflammation of the urinary bladder increases the urinary pH and contributes to the precipitation of struvite crystals that participate in urethral plug formation could be an explanation for this finding.²⁴ Another explanation could be an increase of urinary pH that occurs with a lack of food intake due to the obstruction,²⁵ or due to stress causing hyperventilation and secondary respiratory alkalosis.²⁶

Cats with struvite crystalluria were included in the group of cats with FIC if no other cause of FLUTD could be found, because crystalluria itself normally does not cause clinical signs of FLUTD.²⁷ In a normal feline urinary tract, crystalluria is considered harmless.²² Cats with urethral plugs were also included if no other underlying cause of bladder inflammation could be found. This decision was based on the hypothesis that the leakage of plasma proteins in the urine during inflammation of the urinary bladder from any cause can result in the formation of a urethral plug by precipitated serum proteins.²⁴

An obstruction was significantly more likely in cats with struvite crystalluria, and there was a tendency for struvite crystalluria to be present in a higher number of male cats with an obstruction at presentation ($P = 0.051$). Also, in a study by Kruger et al,¹ crystalluria was significantly more likely in cats with urethral plugs than in controls or cats with other forms of FLUTD. These findings might indicate a role for struvite crystals in the pathogenesis of an obstruction in FIC. However, it is also possible that struvite crystal

formation increases after obstruction due to an increase in urinary pH that can occur with lack of food intake or in stressful situations.^{25,26} Evaluating the urinary crystal status before obstruction occurs might clarify this possibility.

Because most plugs are composed of struvite crystals and a matrix, which is formed by proteins, cells and other material,^{6–8} all of these findings lead to the hypothesis that urethral plugs might be an important cause or contributing factor for obstructions in cats with FIC. Also based on these results, prevention of obstructions by reducing the formation of struvite crystals might be appropriate in obstructed cats with struvite crystalluria. However, further research is necessary to confirm this. Because many factors can influence the formation of struvite crystals in urine, the relevance of struvite crystalluria should not be overinterpreted.^{22,25,26}

Interestingly, urethral plugs were diagnosed in only 37% of obstructed cats, and struvite crystalluria was not present in 33% of obstructed cats. In addition, three male cats with an obstruction at presentation had a low UPC and a low number of red and white blood cells in the urine. One of these cats also did not have crystalluria. Although underdiagnosis of urethral plugs due to flushing back into the urinary bladder during catheterisation is likely, all of these findings indicate that other mechanisms of obstruction (other than urethral plugs) must also be considered in FIC, for example, urethral spasms. Next to surgical complications, urethral spasms might also be the underlying cause of recurring obstructions after perineal urethrostomy in one cat.

Most cats diagnosed with a urethral plug also had struvite crystalluria. However, two cats did not. This illustrates that an obstructive plug can be formed without crystals but that an obstruction is more likely in the presence of crystalluria, as suggested by Gunn-Moore.²⁷

As this was a retrospective case-controlled study, there were inherent limitations.

Determining an appropriate control population was problematic. Although owners of all control cats were thoroughly questioned about the absence of lower urinary tract signs, episodes of FLUTD can go unnoticed. Therefore, cats with past episodes of FLUTD might have been included in the control group. Owners of controls were contacted by phone and no physical examination or urinalysis was performed for control cats at the time of the study. Because of this, cats with FLUTD might have been missed. However, because of the episodic nature of the disease, it was considered unlikely to identify clinically affected cats with a single physical examination and urinalysis. Recall bias was also a concern in this study due to a substantial time lag between diagnosis and the questionnaire in some cats with FIC. Therefore, whenever owners were unsure about specific data, those data were not included in the analysis. Questioning owners about stressful situations that may have occurred several years before might have led to an underestimation of stressful situations in some cases when compared with the controls, who were questioned about

situations during the 3 months immediately prior to the telephone interview.

An important limitation of this study was that urethral stones were not completely excluded as a possible underlying cause given that abdominal radiography was not mandatory in the diagnostic work-up.

In conclusion, several known risk factors for FLUTD were found to play a role in FIC as well. The results support the statement that the combination of a susceptible cat together with a deficient environment is needed for most episodes of FIC to occur. Finally, several findings in this study lead to the hypothesis that urethral plugs might be an important cause or a contributing factor of obstructions in FIC.

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